Linux Fundamentals: User Management Hands-on Assignments

Assignment 1: Basic User Account Provisioning

Production scenario: A new junior developer has joined the application team and needs a standard Linux login account with a home directory and Bash shell.

- Create a user named **devuser01** with a home directory and default shell set to **/bin/bash**.
- Set a temporary password for the user.
- Verify the user account using **id**, **groups**, and the relevant entry in **/etc/passwd**.
- Switch to the new user and create a file named **welcome.txt** inside the home directory.
- Record all commands used and screenshots or terminal output as evidence.

Expected outcome: The student can safely create a regular Linux user, assign a password, confirm UID/GID details, and validate login usability without directly editing system account files.

Assignment 2: User Modification with usermod

Production scenario: An existing employee has moved from a general support role to the application operations team, and the Linux administrator must update the account without deleting or recreating it.

- Create a user named **support01** with a home directory and default Bash shell.
- Modify the user's comment field to include the full name and department, such as **Support User, Application Operations**.
- Change the user's login shell to **/bin/sh**, then verify the change in **/etc/passwd**.
- Create a new home directory path named **/home/appsupport01** and move the existing home directory contents to the new location using the correct user modification option.
- Create a group named **appops** and add **support01** as a supplementary group member without removing any existing group memberships.
- Lock the user account, verify that the account is locked, then unlock it and verify normal access again.

- Document each before-and-after value, including username, UID, home directory, shell, group membership, and lock status.

Expected outcome: The student understands how to safely modify existing Linux user accounts using **usermod**, including updating metadata, shell, home directory, supplementary groups, and lock status while preserving production account continuity.

Assignment 3: User Password and Account Aging Policy

Production scenario: A security team has asked the Linux administrator to enforce password aging and account expiry controls for temporary and permanent users on a production server.

- Create a user named **contractor01** with a home directory and Bash shell.
- View the current password aging settings for **contractor01** using the appropriate command.
- Configure password aging so the user must keep a password for at least **7 days**, must change it after **90 days**, and receives a warning **14 days** before expiry.
- Set the account to expire on a specific future date to simulate a contract end date.
- Configure an inactive period so the account is disabled if the password remains expired for **30 days**.
- Force the user to change the password at the next login and verify the effect.
- Compare the user-specific aging settings with the system-wide defaults in **/etc/login.defs**, and explain whether the defaults apply to existing or only newly created users.
- Prepare production evidence showing the before-and-after aging policy, commands used, and validation output.

Expected outcome: The student understands password maximum age, minimum age, warning period, inactivity lock, account expiry date, forced password change, and how these controls support production security and compliance requirements.